

Nau mai, welcome to

Applied Engineering Optimisation

ENEL445

Week 1 Lecture 2

- How to formulate optimisation problems
 - Design variables
 - Objective (cost) function
 - Constraints
- Optimisation problem classification
- Mathematical background

Design (decision) variables

- These are the variables that we can change to achieve our desired result:
- We usually denote them as follows:
- When setting up an optimization problem, a good starting point is to think “what variables do I have control over”?

Design variables

- Design variables can be continuous or discrete, or a mixture.
- Design variables should be independent, otherwise there may be an infinite number of equivalent solutions.
- Don't forget to add physically meaningful bounds.
- Tip: when starting to solve a problem, try to use as few design variables as possible.

Objective function

- This is the objective by which we measure the quality of a solution.
- In this course, we will usually have a closed-form function $f(x)$ which serves as the objective to be minimised or maximised.
- However, in many applications, the objective function could be the output of a simulation or physical system/experiment.

Objective function

- We will look at minimisation problems in this course. If you have a maximization problem, you can always simply reverse the sign of the objective function!
- The choice of objective function is crucial for successful design optimization. If this is done poorly, the mathematical optimum will be non-optimal from the engineering point of view.
- A bad choice for the objective function is a common mistake in design optimization.

Choosing an objective function

- Example: designing a car for Toyota
- What should your objective function be?
 - a. The weight of the car in kg (lower is better).
 - b. The manufacturing cost of the car in \$.
 - c. The performance (speed, acceleration, handling, torque, etc.) of the car.
 - d. The capacity (people and baggage) of the car.
 - e. Some weighted combination of some of the above.

Choosing an objective function

- Example: going shopping at the supermarket
- What should your objective function be?
 - a. The total cost of your groceries.
 - b. Meeting your nutritional requirements as closely as possible.
 - c. Required preparation time (minimise).
 - d. How much you like the food (maximise)
 - e. Some weighted combination of some of the above.

Objective functions

- As we can see, many real-life problems are complicated, and the objective function is a weighted combination of various considerations.
- Alternatively, **multi-objective optimisation** can be performed directly.

Constraints

- Many optimisation problems involve constraints:
 - E.g. when designing a car you are limited by the available parts, by road rules governing size, by safety requirements etc.
 - When grocery shopping you are limited by stock availability, budget limits, item limits, etc.
- There are two types of constraints:
 - Inequality constraints
 - Equality constraints (equivalent to two inequality constraints)

Constraints

- Inequality constraints can be active or inactive at the optimum point. An active inequality constraint means that $g(x^*) = 0$, whereas for an inactive one, $g(x^*) < 0$.

Constraints

- The region in which all constraints are satisfied is called the *feasible region*.
- If there are too many constraints, there may be no feasible region and this makes the problem also *infeasible*.
- Example: if there are more independent equality constraints than design variables, the problem is infeasible.

Constraints vs Objective function terms

- Example: going shopping at the supermarket
- Which ones could be constraints?
 - a. The total cost of your groceries (e.g. total cost less than \$100)
 - b. Meeting your nutritional requirements (e.g. calories $\geq X$, fat $\leq Y$, etc.)
 - c. Required preparation time (e.g. time/day ≤ 1 hr)
 - d. How much you like the food (e.g. all foods below a threshold of liability are out)

Putting it all together (+notation)

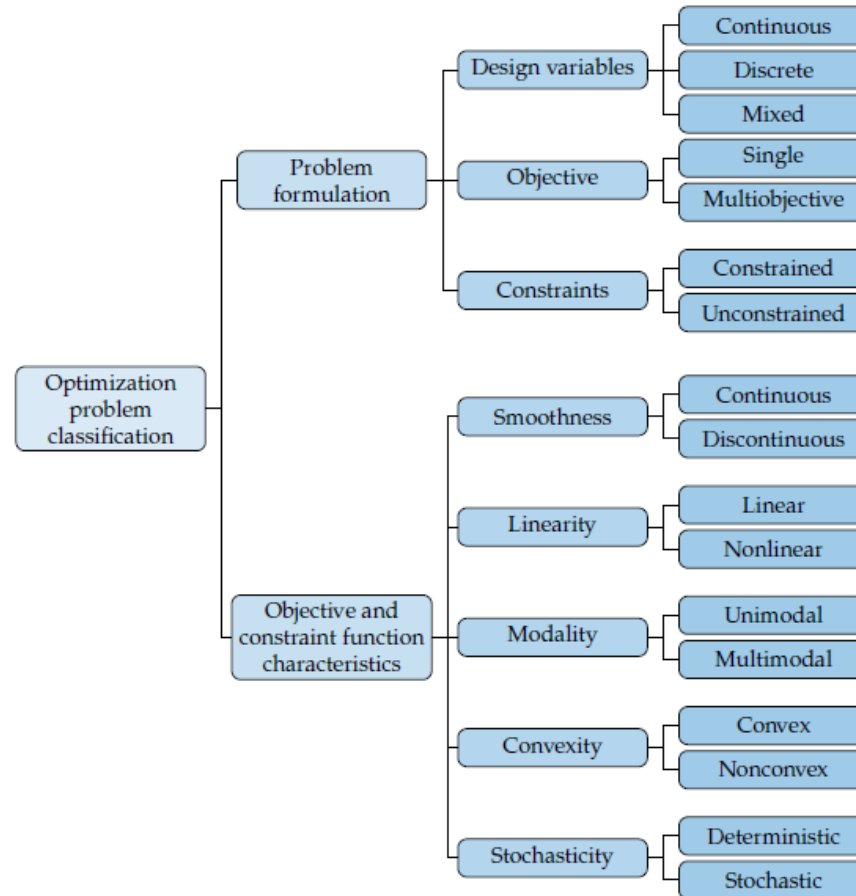
$$\min f(\mathbf{x})$$

$$\text{subject to } \underline{x}_i \leq x_i \leq \bar{x}_i, i = 1, 2, \dots, n$$

$$g_j(\mathbf{x}) \leq 0, \quad j = 1, 2, \dots, n_g$$

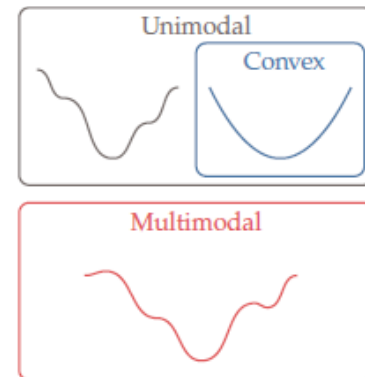
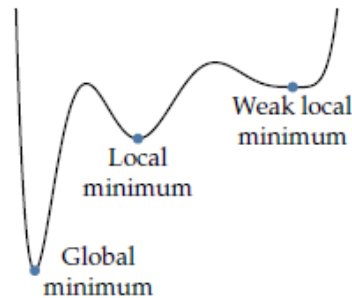
$$h_l(\mathbf{x}) = 0, \quad l = 1, 2, \dots, n_h.$$

Optimization classification



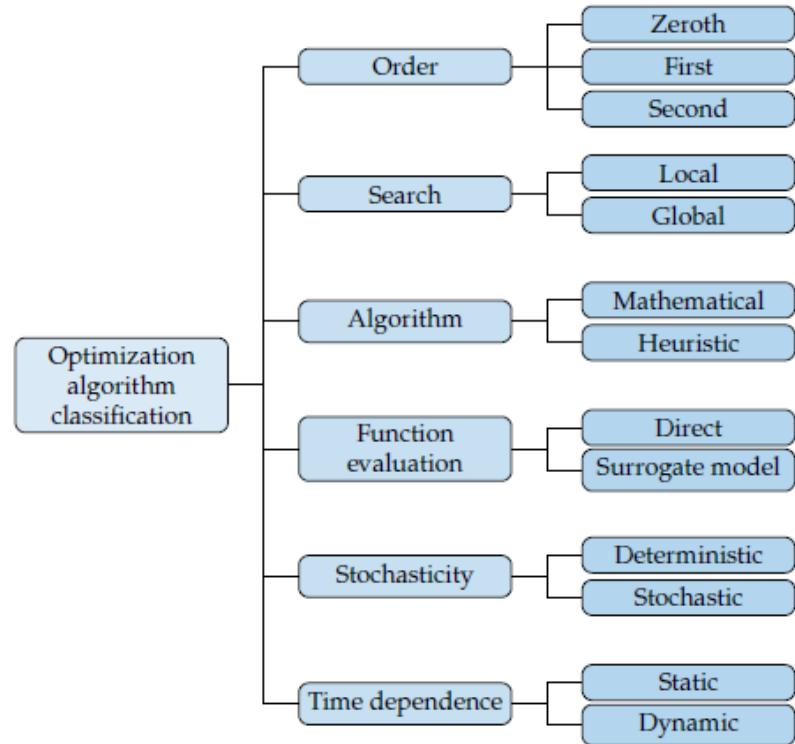
Mathematical concepts

- Smooth / continuous functions
- Linearity
- Multimodal vs unimodal

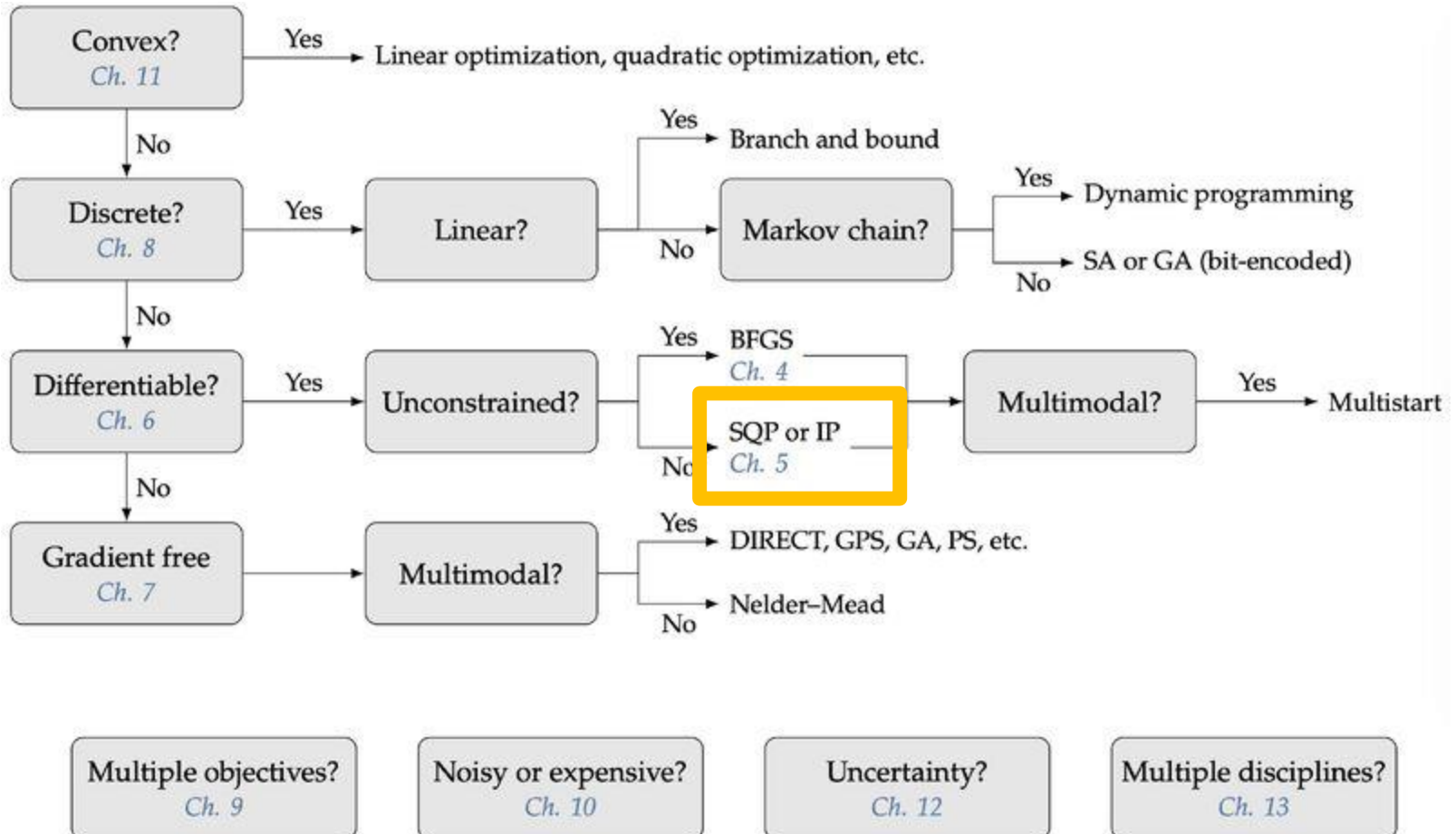


Optimisation algorithms

- No optimization algorithm is good for every problem!
- Gradient vs gradient-free optimisation



Which optimiser should I use?



Notation

- Vectors
- Matrices and elements of matrices
- Optimum point x^*

Vector and matrix norms

Vector and matrix math concepts

trace

symmetric matrices

positive (negative) (semi-)definiteness

gradient

Hessian

Vector derivatives

Taylor-series expansion